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Abstract

Water is one of the most essential natural resources for sustaining human life, agriculture, industries, and aquatic ecosystems. The quality of water directly influences public health, food production, environmental sustainability, and the survival of aquatic organisms. Although water may appear clean to the naked eye, it may still contain unsafe pH levels or experience temperature changes that make it unsuitable for drinking, household use, or fish farming. Conventional methods of testing water quality generally involve laboratory analysis or manual testing kits, which are often expensive, time-consuming, and unable to provide continuous monitoring. Therefore, there is a growing need for an affordable and automated system that can monitor water quality in real time.

This project presents a **Water Quality Monitoring System Using Arduino**, designed to continuously monitor two important water quality parameters: **pH level** and **water temperature**. The system is built using an **Arduino Uno**, a **pH Sensor Module**, a **DS18B20 waterproof temperature sensor**, an **active buzzer**, and a few supporting electronic components. The pH sensor measures the acidity or alkalinity of water, while the temperature sensor measures the current water temperature. The Arduino collects and processes the sensor data, compares the readings with predefined safe limits, and activates the buzzer whenever the values exceed the acceptable range. The measured values can also be viewed in real time through the Arduino IDE Serial Monitor.

The proposed system is simple, portable, energy-efficient, and cost-effective, with a total hardware cost of approximately **₹3785**. It can be used in households, aquariums, fish farms, educational laboratories, and small-scale water treatment systems. In addition to monitoring water quality, the project provides practical experience in embedded systems, sensor interfacing, Arduino programming, and real-time monitoring technology. Although the current prototype measures only pH and temperature, it has been designed with future expansion in mind. Additional sensors, wireless communication modules, cloud storage, and mobile applications can easily be integrated to transform it into a complete IoT-based smart water quality monitoring system. Therefore, this project serves as an effective, affordable, and scalable solution for continuous water quality monitoring while providing a strong foundation for future research and development.

Keywords: Arduino Uno, Water Quality Monitoring, pH Sensor, DS18B20 Temperature Sensor, Embedded System, Smart Monitoring, Internet of Things (IoT).

1. Project Title:

Water Quality Monitoring System Using Arduino

1.1 Introduction:

Water is one of the most valuable natural resources on Earth and plays a vital role in human health, agriculture, industrial production, and aquatic ecosystems. Safe and clean water is essential for drinking, cooking, farming, fish cultivation, and many industrial applications. However, water quality can change due to pollution, chemical contamination, environmental conditions, or natural processes. These changes are not always visible, which means water may appear clean while still being unsafe for use. Among the many factors that determine water quality, **pH level** and **temperature** are two of the most important. An improper pH level can make water acidic or alkaline, affecting human health, damaging pipelines, and harming aquatic life. Similarly, abnormal water temperature can reduce dissolved oxygen levels, negatively affecting fish and other aquatic organisms.

Traditional methods of monitoring water quality usually involve manual testing using chemical strips or laboratory equipment. Although these methods provide accurate results, they require trained personnel, take considerable time, and cannot provide continuous monitoring. As a result, sudden changes in water quality may remain undetected until serious damage has already occurred. For example, a sudden change in water temperature or pH in a fish pond can result in fish mortality before manual testing is performed. Therefore, an automated and real-time monitoring system is highly desirable.

The **Water Quality Monitoring System Using Arduino** has been developed to solve this problem by providing a simple, low-cost, and efficient solution. The system uses an **Arduino Uno** as the main controller, a **pH Sensor Module** to measure the acidity or alkalinity of water, and a **DS18B20 waterproof temperature sensor** to measure water temperature. These sensors continuously send data to the Arduino, which compares the measured values with predefined safe limits. Whenever the readings exceed the acceptable range, an **active buzzer** immediately alerts the user so that corrective action can be taken. The measured values are also displayed through the Arduino Serial Monitor, allowing users to observe water conditions in real time.

One of the main advantages of this project is its affordability and simplicity. The total hardware cost is approximately **₹3785**, making it accessible to students, researchers, homeowners, aquarium owners, and small-scale fish farmers. Furthermore, the system has a modular design, allowing additional features such as Wi-Fi connectivity, cloud data storage, mobile application support, LCD displays, and extra water quality sensors to be added in the future. These enhancements can transform the project into a complete IoT-based smart water monitoring system. Overall, this project demonstrates how embedded systems and sensor technology can be used to develop an efficient, reliable, and scalable solution for continuous water quality monitoring while providing valuable educational experience in electronics, programming, and environmental monitoring.

2. Purpose:

2.1 Objective:

The primary objective of this project is to develop a simple, affordable, and reliable water quality monitoring system using Arduino technology. The system continuously measures water pH and temperature in real time and provides an immediate warning whenever unsafe conditions are detected.

The specific objectives of this project are:

- To monitor the pH level of water continuously.
- To measure water temperature accurately using a waterproof sensor.
- To detect unsafe water conditions automatically.
- To alert users instantly using an active buzzer.
- To reduce the need for manual water testing.
- To provide an affordable solution for households, fish farms, aquariums, and educational institutions.
- To develop a practical embedded system using Arduino.
- To create a platform that can be expanded into an IoT-based smart monitoring system in the future.

2.2 Scope:

The proposed Water Quality Monitoring System has a wide range of applications due to its low cost, portability, and ease of operation.

The system can be used in:

- ✓ Household drinking water storage tanks
- ✓ Fish ponds and commercial fish farms
- ✓ Home and commercial aquariums
- ✓ Schools, colleges, and university laboratories
- ✓ Small-scale water treatment plants
- ✓ Agricultural irrigation systems
- ✓ Environmental monitoring projects
- ✓ Research laboratories

The current version of the system measures only pH and water temperature while providing an audible alert through a buzzer. However, the project has been designed in a modular way so that future improvements can be easily added.

Possible future expansions include:

- Wi-Fi monitoring using ESP32
- Bluetooth connectivity
- LCD or OLED display
- Mobile application
- Cloud database
- SMS notification
- Water level sensor
- Turbidity sensor
- TDS sensor
- Dissolved Oxygen (DO) sensor

These improvements would transform the prototype into a complete smart IoT-based water quality monitoring system suitable for industrial and commercial applications.

2.3 Significance:

Water quality is essential for human health, agriculture, and aquatic life. Checking water quality manually using test strips or laboratory tests takes time, requires technical knowledge, and cannot provide continuous monitoring. This project solves that problem by automatically monitoring the water's pH level and temperature in real time. Whenever the water becomes unsafe, the system immediately alerts the user through a buzzer. With a total hardware cost of approximately \$3785, the system is affordable for households, students, fish farmers, and small-scale users. It provides a simple, reliable, and cost-effective solution for everyday water quality monitoring.

2.4 Motivation Behind the Project:

The motivation behind this project is to develop a simple, affordable, and reliable system for continuous water quality monitoring. Water may appear clean but still have an unsafe pH level or temperature, which cannot be detected by visual inspection. This project provides real-time monitoring and an instant buzzer alert, helping users take quick action before serious problems occur.

The project was developed to address the following needs:

- **Ensure Water Safety:** Continuously monitor water quality and detect unsafe conditions in real time.
- **Reduce Manual Testing:** Eliminate the need for frequent testing using pH strips or laboratory methods.
- **Protect Aquatic Life:** Help prevent fish deaths caused by sudden changes in pH level or water temperature.
- **Provide Early Warning:** Alert users immediately through a buzzer when water quality becomes unsafe.
- **Offer a Low-Cost Solution:** Develop an affordable system suitable for households, fish farms, aquariums, and educational laboratories.

2.5 Applications of the Proposed System:

This project is most useful for the following tasks:

- Early warning: Alerting the user immediately when water becomes unsafe, instead of waiting for a scheduled manual check.
- Protecting fish and aquatic life: Helping fish farmers and aquarium owners avoid sudden fish deaths caused by unsafe pH or temperature.
- Checking drinking water safety: Giving households a simple way to know if their stored water is within a safe pH range before use.
- Reducing manual work: Removing the need for someone to manually test the water every day using strips or chemical kits.
- Supporting learning and research: Giving students an easy, low-cost way to practice real-time sensor data collection and embedded systems in school or college projects.
- Building a foundation for bigger systems: Acting as the first working version of a larger, smarter water-monitoring system that can later add more sensors, alerts, or app connections.

3. Components:

The following hardware components were used to develop the **Water Quality Monitoring System Using Arduino**. Each component plays an important role in collecting sensor data, processing information, and providing alerts to the user.

Component	Quantity	Description
Arduino Uno (R3)	1	This is the "brain" of the project. It reads data from the sensors, checks if the values are safe, and turns the buzzer on if needed.
pH Sensor Module (with Probe)	1	This part goes into the water and measures how acidic or basic (alkaline) the water is. It sends this reading to the Arduino.
DS18B20 Waterproof Temperature Sensor	1	This part measures the water temperature and sends the reading to the Arduino.
Active Buzzer (5V)	1	This makes a beeping sound when the water is not safe, so the user knows something is wrong
Breadboard	1	A board used to connect all the wires and parts together without needing to solder anything.
4.7kΩ Resistor	1	A small part needed for the temperature sensor to send its data correctly. It is called a "pull-up resistor."
Jumper Wires	1 Set	Used to connect all components together.
USB Cable	1	Used to give power to the Arduino and to upload the program (code) from a computer.

4. Diagram/Circuit Setup:

The circuit is designed using an Arduino Uno, a pH sensor module, a DS18B20 waterproof temperature sensor, an active buzzer, and a breadboard. The Arduino supplies power to all the components and continuously reads sensor data to monitor water quality.

Here is how everything is connected:

- The Arduino's 5V pin and GND pin are connected to the breadboard's power rows, so all parts can share power and ground.
- The pH sensor's power and ground wires go to the breadboard's power rows. Its signal wire goes to Arduino pin A0.
- The temperature sensor's power and ground wires go to the breadboard's power rows. Its data wire goes to Arduino pin D2. The 4.7k ohm resistor is placed between the data wire and the 5V line — this helps the sensor send correct readings.
- The buzzer's plus (+) wire goes to Arduino pin D8, and its minus (-) wire goes to GND.

Quick connection guide:

Part	Connect To Arduino	Notes
pH Sensor Module	Signal wire -> A0	Power wire -> 5V, Ground wire -> GND
Temperature Sensor (DS18B20)	Data wire -> D2	Power -> 5V, Ground -> GND. Put the 4.7k ohm resistor between the Data wire and 5V.
Buzzer	Plus (+) wire -> D8	Minus (-) wire -> GND

How it works: The Arduino keeps reading the pH sensor and changes the reading into a pH number (0 to 14) using a simple formula. At the same time, it asks the temperature sensor for the current water temperature. If the pH number is outside the safe range (about 6.5 to 8.5) or the temperature is too high, the Arduino turns the buzzer on to warn the user. The readings can also be seen on a computer screen using the Arduino Serial Monitor.

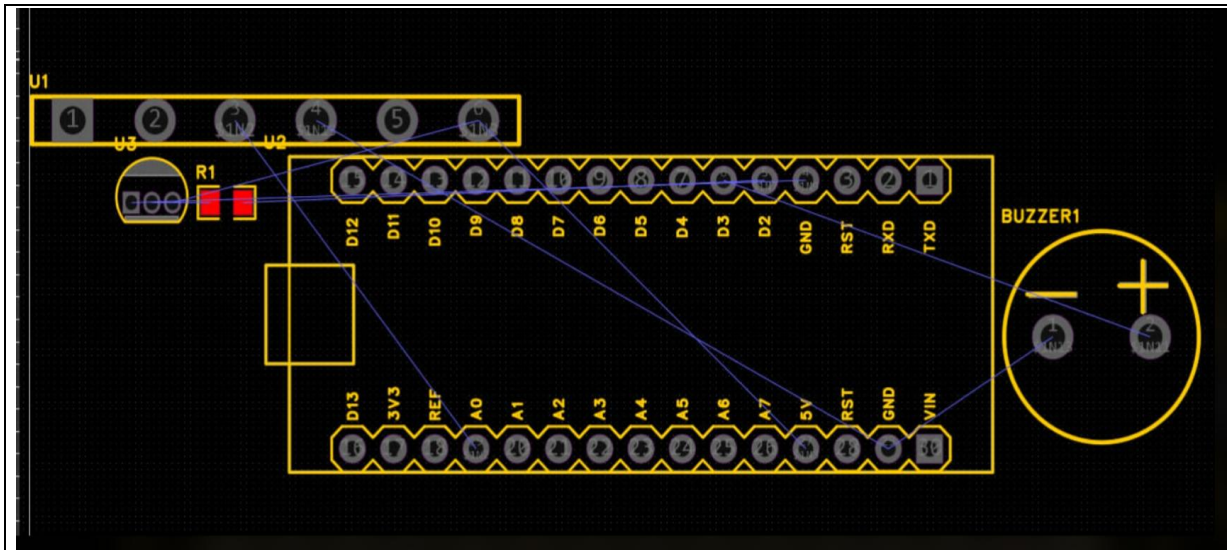


Figure 4.1: PCB Layout of the Water Quality Monitoring System

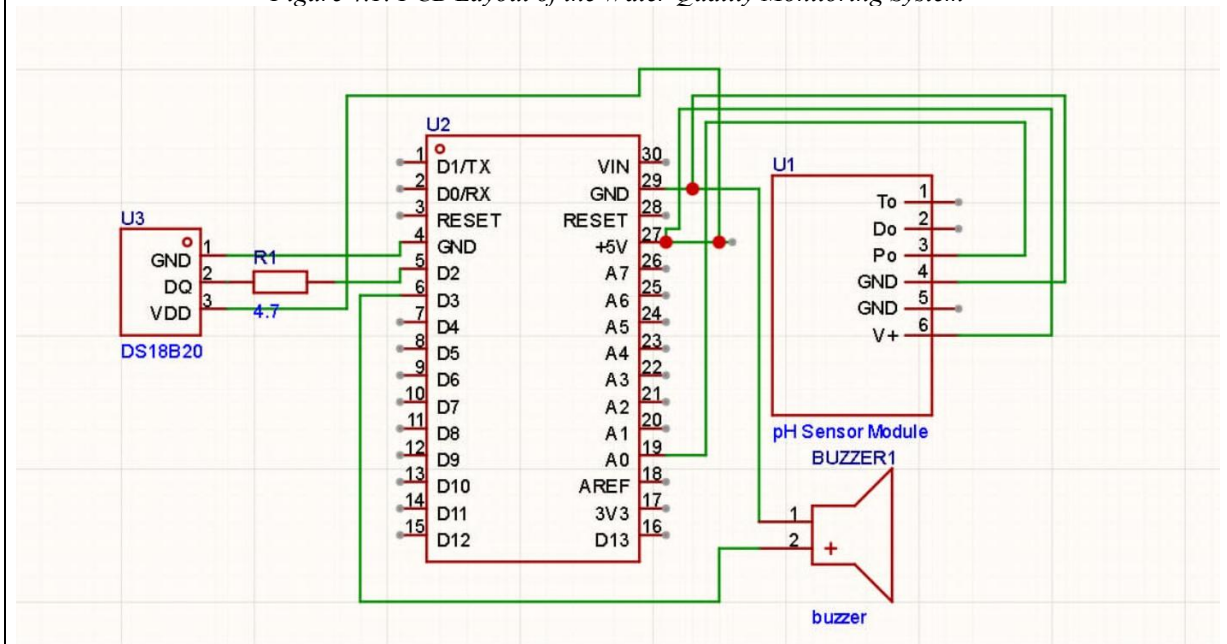


Figure 4.1: PCB Design of the Water Quality Monitoring System

5. Cost Breakdown:

Component	Quantity	Approx. Cost
Arduino Uno (R3)	1 pc	₹850
pH Sensor Module (with probe)	1 pc	₹2500
Temperature Sensor (DS18B20, waterproof)	1 pc	₹70
Buzzer (Active, 5V)	1 pc	₹40

Component	Quantity	Approx. Cost
Breadboard (830 point)	1 pc	₹100
Resistor (4.7k ohm)	1 pc	₹5
Jumper Wires (Male-Male / Male-Female)	1 set (20 pcs)	₹120
USB Cable (to power Arduino)	1 pc	₹100
Total		₹3785 (approx.)

(Prices can change a little depending on the shop and market.)

The total hardware cost of the project is approximately **₹3785**, making it an affordable solution for students, researchers, households, and small-scale users. Since all components are readily available in local electronics markets, the system is easy to build, maintain, and upgrade.

6. Functionality Breakdown:

6.1 Functionality 1: pH Monitoring

The pH sensor stays in the water and keeps sending a reading to the Arduino. The Arduino turns this reading into a pH value using a formula that is set up with known test solutions (like pH 4, 7, and 10). If the pH value goes outside the safe range (normally 6.5 to 8.5 for drinking water and most fish), the system marks the water as "not safe" and gets ready to warn the user.

6.2 Functionality 2: Temperature Monitoring & Alert System

The temperature sensor keeps checking the water temperature and sends this reading to the Arduino. The Arduino compares it with a safe temperature range that is already set in the program. If the temperature is too high or too low, or if the pH value is also unsafe, the Arduino turns on the buzzer right away to alert the user. Once the readings go back to a safe range, the buzzer turns off automatically.

7. Business Proposal:

This project can be sold as an affordable water-checking device for fish farmers, small water suppliers, aquarium owners, and schools. Since the parts cost less than ₹2500 in total, the finished product can be sold at a good price and still leave a fair profit.

Potential business opportunities include:

- Selling ready-made water quality monitoring devices to households and fish farmers.
- Providing customized monitoring systems for aquariums and research laboratories.
- Offering installation and maintenance services.
- Developing IoT-based premium versions with mobile applications and cloud monitoring.

- Providing freelance hardware development services through platforms such as Fiverr and Upwork.

Some ways to grow this into a bigger business:

- Add a Wi-Fi or Bluetooth module (like ESP8266/ESP32) so users can check the water reading from their phone. This can later become a paid monitoring service (subscription).
- Add a memory card or online storage to save past readings, so users can see how water quality has changed over time — useful for fish farms that need to keep records.
- Offer this as a custom project for clients on freelance sites like Upwork or Fiverr, for anyone who wants their own water-monitoring setup.
- Add more sensors later (like turbidity or TDS sensors) as extra paid features for more complete water testing.

8. Potential Challenges:

- Sensor calibration: The pH sensor reading can slowly change over time, so it needs to be re-calibrated now and then with standard test solutions.
- Wear and tear: Keeping the sensors in water for a long time can slowly damage or wear out the probe.
- Noisy readings: The pH sensor's readings can sometimes jump around a little, so the code needs to take an average of several readings to get a stable value.
- Power cuts: If the power goes off, the whole system stops working. A backup power source (like a power bank) should be added for important places.
- Weatherproofing: If the device is used outdoors (like a pond), the Arduino and wires need a waterproof box, since only the sensor tips are made to touch water.
- Different safe ranges: Safe pH and temperature levels are different for drinking water, fish ponds, and aquariums, so these values should be easy to change in the code rather than fixed.

9. Conclusion:

The Water Quality Monitoring System Using Arduino is a simple, affordable, and reliable solution for monitoring water quality in real time. By continuously measuring pH and temperature and providing an instant buzzer alert, the system helps users detect unsafe water conditions quickly. With future enhancements such as IoT connectivity and additional sensors, it can be developed into a complete smart water quality monitoring system for both domestic and commercial use.